Congratulations! You passed!

TO PASS 90% or higher

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Practice Quiz

TOTAL POINTS 15

1. TD(0) is a solution method for: 1 / 1 point

- ○ Control
- ◉ Prediction

Correct
Correct! TD(0) is used to estimate the value function for a given policy. In other words, it is a solution method for the prediction problem.

2. Which of the following methods use bootstrapping? (Select all that apply) 2 / 2 points

- [x] Dynamic Programming

Correct
Correct! DP algorithms are obtained by turning Bellman equations into assignments, that is, into update rules for improving approximations of the desired value functions. These methods update estimates of the values of states based on estimates of the values of successor states. That is, they update estimates on the basis of other estimates.

- [x] TdB0

Correct
Correct! Temporal Difference methods update "a guess from a guess". They estimate the value of the current state using the immediate reward and the estimate of the value in the next state. They bootstrap off their own estimates.

3. Which of the following is the correct characterization of Dynamic Programming (DP) and Temporal Difference (TD) methods? 1 / 1 point

- ○ Both TD methods and DP methods require a model: the dynamics function p.
- ○ Neither TD methods nor DP methods require a model: the dynamics function p.
- ○ TD methods require a model, the dynamics function p, but Monte-Carlo methods do not.
- ◉ DP methods require a model, the dynamics function p, but TD methods do not.

Correct
Correct! Dynamic Programming methods solve Bellman equations using a model. TD methods use sample updates from the environment, and do not need to explicitly have the dynamics function p.

4. Which of the following correctly pairs a prediction algorithm and an update? 2 / 2 points

- [ ] Monte Carlo: $V(S_t) \leftarrow V(S_t) + \alpha(R_t + \gamma V(S_{t+1}) - V(S_t))$
- [x] Monte Carlo: $V(S_t) \leftarrow V(S_t) + \alpha(G_t - V(S_t))$

,

- [x] TDdB: $V(S_t) \leftarrow V(S_t) + \alpha(R_t + \gamma V(S_{t+1}) - V(S_t))$

Correct
Correct! TD(0) updates value estimates toward the TD(0) target of the sum of the observed reward and discounted next state value.

- [ ] TDdB: $V(S_t) \leftarrow V(S_t) + \alpha(G_t - V(S_t))$

5. Which of the following well-describe Temporal Difference (TD) and Monte-Carlo (MC) methods? 2 / 2 points

- [x] TD methods are used in continuing tasks.

Correct
Correct! The returns in continuing tasks are sums of rewards infinitely into the future. But, TD does not have to wait to get samples of these returns. The targets can be obtained immediately, using bootstrapping.

- [ ] MC methods are used in continuing tasks.
- [x] TD methods are used in episodic tasks.

Correct
Correct! TD updates on every step, using bootstrapped targets. This means it can be used in continuing and episodic tasks.

- [x] MC methods are used in episodic tasks.

Correct

6. Suppose we have current estimates for the value of two states: V(A) = 1.0, V(B) = 1.0 in an episodic setting. We observe the following trajectory: A, 0, B, 1, B, 1, terminal. Assume T is a terminal state. Apply TD(0) with step-size, $\alpha = 1$, and discount factor, $\gamma = 0.5$. What are the value estimates for state A and state B at the end of the episode? Provide your answers to 1 decimal place in the following format, replacing V(A) and V(B) with your answers: (V(A), V(B)) 4 / 4 points

(0.5, 1)

Correct
Correct! The steps to the answer are presented below:

After observing A, A, B:

$V(A) \leftarrow V(A) + \alpha \cdot [R + \gamma V(B) - V(A)]$

Simplifying, $V(A) = 1.0 + 1 \cdot [0 + 0.5 \cdot 1.0 - 1.0]$.

So, $V(A) = 1 + (-0.5)$. Thus, $V(A) = 0.5$.

$V(B)$ remains the same.

Therefore, after this transition, $V(A) = 0.5, V(B) = 1$.

After observing B, 1, B:

$V(B) \leftarrow V(B) + \alpha \cdot [R + \gamma V(B) - V(B)]$

Simplifying, $V(B) = 1 + 1 \cdot [1 + 0.5 \cdot 1 - 1]$.

So, $V(B) = 1 + [0.5]$. Thus, $V(B) = 1.5$.

After observing B, 0, T:

$V(B) \leftarrow V(B) + \alpha \cdot [R + \gamma V(T) - V(B)]$

Simplifying, $V(B) = 1.5 + 1 \cdot [0 + 0.5 \cdot 0 - 1.5]$ and

$V(B) = 1.5 + [-1.5]$. Thus, $V(B) = 0$.

$V(A)$ remains the same.

Therefore, after this transition: $V(A) = 0.5, V(B) = 0$.

Thus the answer is (0.5, 0.0).

7. Which of the following pairs is the correct characterization of the targets used in TD(0) and Monte Carlo? 1 / 1 point

- ○ TD(0): High Variance Target, Monte Carlo: High Variance Target
- ○ TD(0): High Variance Target, Monte Carlo: Low Variance Target
- ◉ TD(0): Low Variance Target, Monte Carlo: High Variance Target
- ○ TD(0): Low Variance Target, Monte Carlo: Low Variance Target

Correct
Correct! MC targets generally have higher variance while TD(0) targets usually have lower variance.

episodes
A, 0, B, 0
B, 1
B, 1
B, 1
B, 0
B, 0
B, 1
B, 0

What would batch Monte Carlo methods give for the estimates V(A) and V(B)? What would batch TD(0) give for the estimates V(A) and V(B)?

For Batch MC, compute the average returns observed from each state. For Batch TD: You can start with state B. What is its expected return? Then figure out V(A) using the temporal difference equation: $V(S_t) \leftarrow E[R_{t+1} + \gamma V(S_{t+1})]$.

Provide your answers to 1 decimal place in the following format: $(V^{\text{batch MC}}(A), V^{\text{batch MC}}(B), V^{\text{batch TD}}(A), V^{\text{batch TD}}(B))$ **and replace**

- $V^{\text{batch MC}}(A)$,
- $V^{\text{batch MC}}(B)$,
- $V^{\text{batch TD}}(A)$, **and**
- $V^{\text{batch TD}}(B)$

with your answers.

Correct
Correct! See Section 6.3 and Example 6.4 of the textbook for more details.

9. True or False: "Both TD(0) and Monte-Carlo (MC) methods converge to the true value function asymptotically, given that the environment is Markovian." 1 / 1 point

- ◉ True
- ○ False

Correct
Correct! See Section 6.2, "Advantages of TD Prediction methods" in the book.

10. Choose the correct algorithm, backup diagram pairs: 2 / 2 points

- [x] TDdB:

reward and new state wmch togetner provide the r_t target.

- [ ] Monte-Carlo:

- [ ] TDdB:

- [x] Monte-Carlo:

Correct
Correct! See Page 95 of the textbook. This backup diagram corresponds to the Monte-Carlo algorithm where an agent updates a state value based on the empirically observed return, seen at the end of the episode.

11. Which of the following pairs is the correct characterization of the TD(0) and Monte-Carlo (MC) methods? 1 / 1 point

- ○ Both TD(0) and MC are offline methods.
- ○ Both TD(0) and MC are online methods.
- ◉ TD(0) is an online method while MC is an offline method.
- ○ MC is an online method while TD(0) is an offline method.

Correct
Correct! A primary advantage of TD(0) is that it can learn during an episode: it can learn online